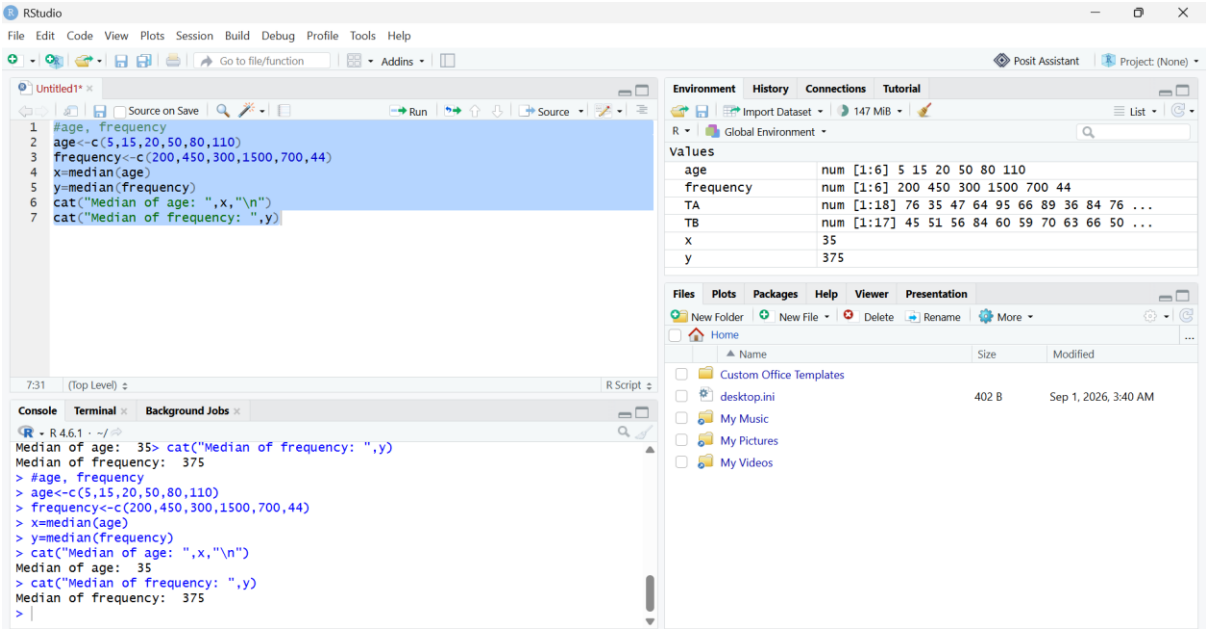
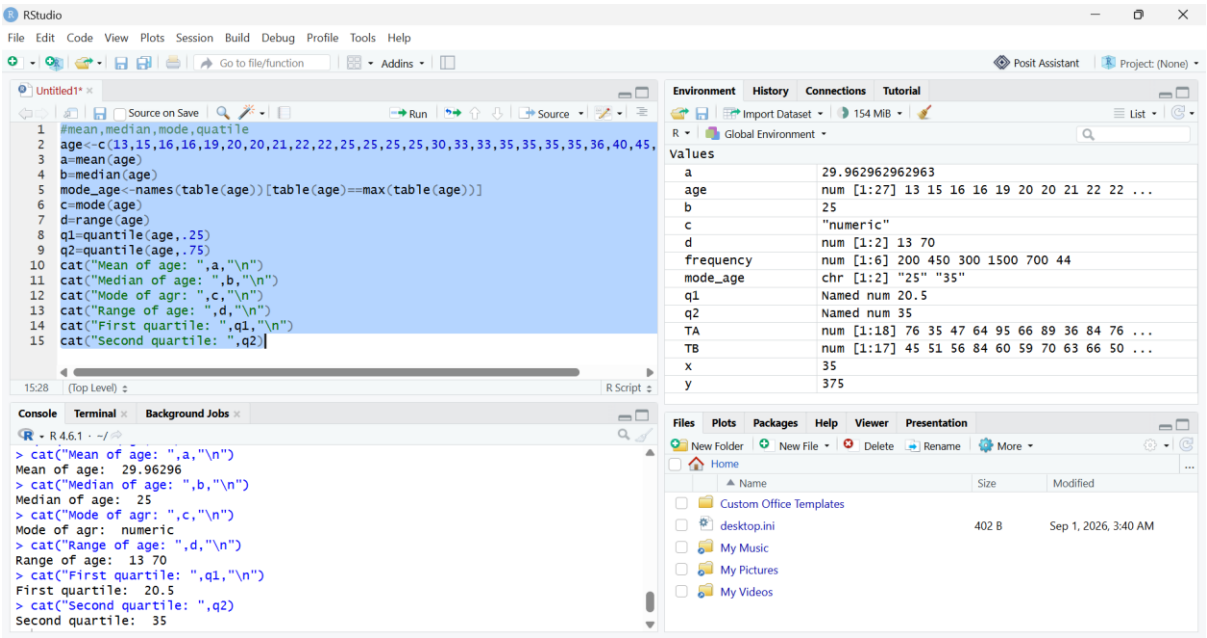
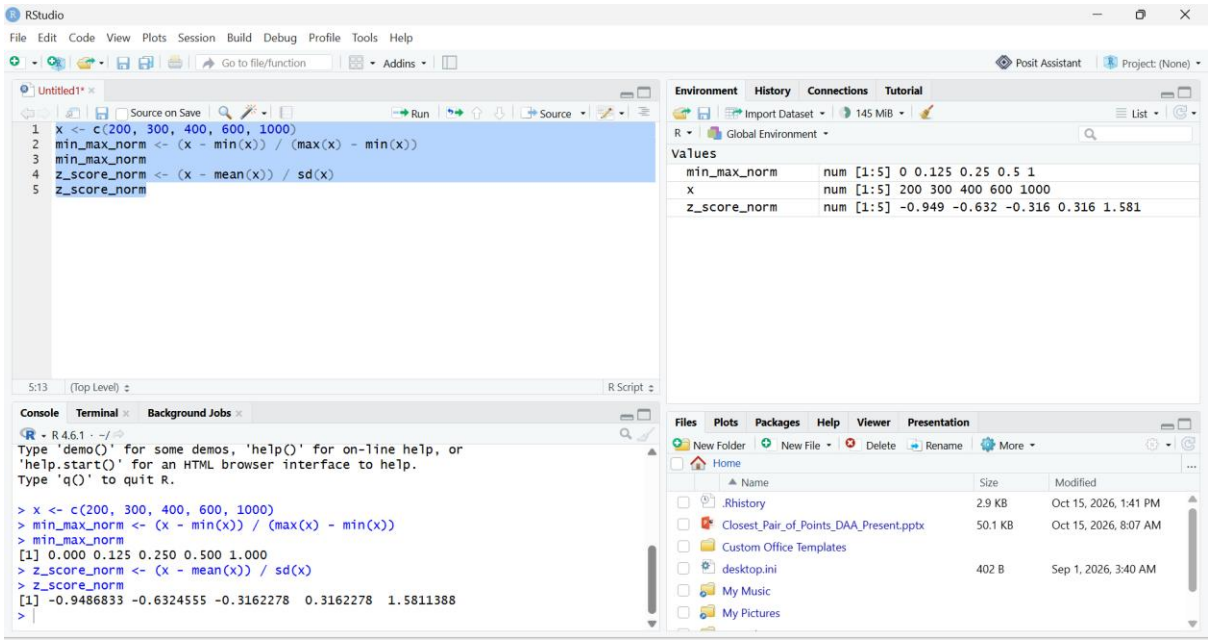
Q1.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
Source on Save
Run
Source
#age, frequency
age<-c(5,15,20,50,80,110)
frequency<-c(200,450,300,1500,700,44)
x=median(age)
y=median(frequency)
cat("Median of age: ",x,"\n")
cat("Median of frequency: ",y)
7:31 (Top Level)
R Script
Console Terminal Background Jobs
R 4.6.1 · ~/
Median of age: 35> cat("Median of frequency: ",y)
Median of frequency: 375
> #age, frequency
> age<-c(5,15,20,50,80,110)
> frequency<-c(200,450,300,1500,700,44)
> x=median(age)
> y=median(frequency)
> cat("Median of age: ",x,"\n")
Median of age: 35
> cat("Median of frequency: ",y)
Median of frequency: 375
Environment History Connections Tutorial
Import Dataset
147 MiB
Global Environment
Values
age num [1:6] 5 15 20 50 80 110
frequency num [1:6] 200 450 300 1500 700 44
TA num [1:18] 76 35 47 64 95 66 89 36 84 76 ...
TB num [1:17] 45 51 56 84 60 59 70 63 66 50 ...
x 35
y 375
Files Plots Packages Help Viewer Presentation
New Folder
New File
Delete
Rename
More
Home
Name
Size
Modified
Custom Office Templates
desktop.ini 402 B Sep 1, 2026, 3:40 AM
My Music
My Pictures
My Videos

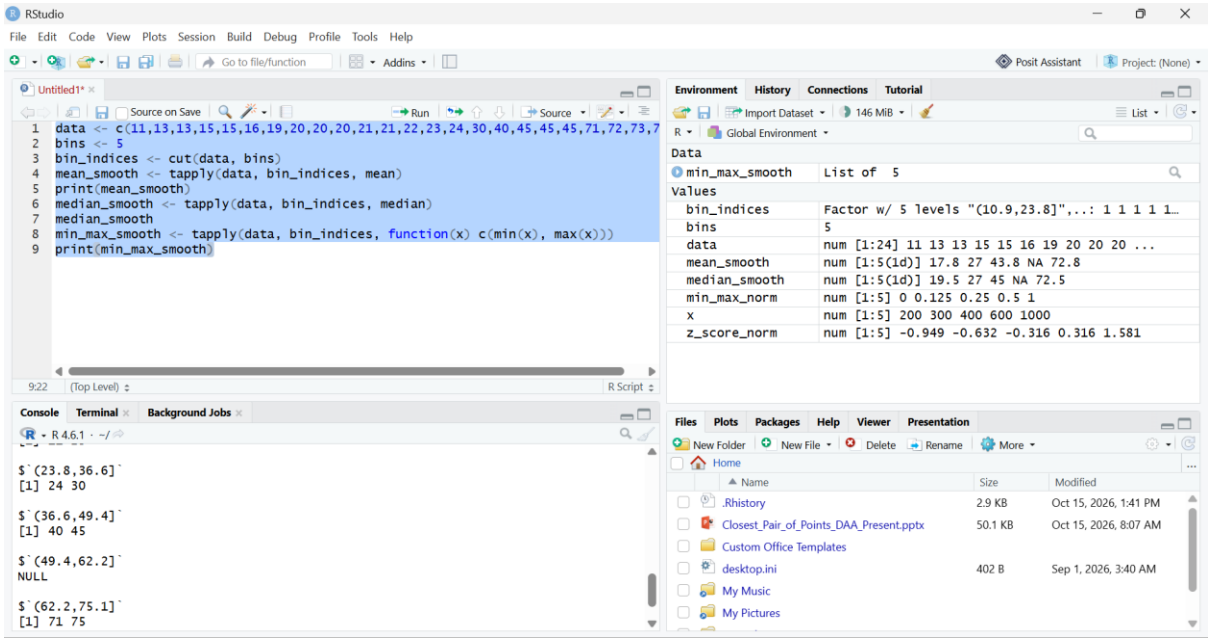
Q2.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
Source on Save
Run
Source
#mean,median,mode,quatile
age<-c(13,15,16,16,19,20,20,21,22,22,25,25,25,30,33,33,35,35,35,36,40,45,
a=mean(age)
b=median(age)
mode_age<-names(table(age))[table(age)==max(table(age))]
c=mode(age)
d=range(age)
q1=quantile(age,.25)
q2=quantile(age,.75)
cat("Mean of age: ",a,"\n")
cat("Median of age: ",b,"\n")
cat("Mode of agr: ",c,"\n")
cat("Range of age: ",d,"\n")
cat("First quartile: ",q1,"\n")
cat("Second quartile: ",q2)
15:28 (Top Level)
R Script
Console Terminal Background Jobs
R 4.6.1 · ~/
> cat("Mean of age: ",a,"\n")
Mean of age: 29.96296
> cat("Median of age: ",b,"\n")
Median of age: 25
> cat("Mode of agr: ",c,"\n")
Mode of agr: numeric
> cat("Range of age: ",d,"\n")
Range of age: 13 70
> cat("First quartile: ",q1,"\n")
First quartile: 20.5
> cat("Second quartile: ",q2)
Second quartile: 35
Environment History Connections Tutorial
Import Dataset
154 MiB
Global Environment
Values
a 29.962962962963
age num [1:27] 13 15 16 16 19 20 20 21 22 22 ...
b 25
c "numeric"
d num [1:2] 13 70
frequency num [1:6] 200 450 300 1500 700 44
mode_age chr [1:2] "25" "35"
q1 Named num 20.5
q2 Named num 35
TA num [1:18] 76 35 47 64 95 66 89 36 84 76 ...
TB num [1:17] 45 51 56 84 60 59 70 63 66 50 ...
x 35
y 375
Files Plots Packages Help Viewer Presentation
New Folder
New File
Delete
Rename
More
Home
Name
Size
Modified
Custom Office Templates
desktop.ini 402 B Sep 1, 2026, 3:40 AM
My Music
My Pictures
My Videos

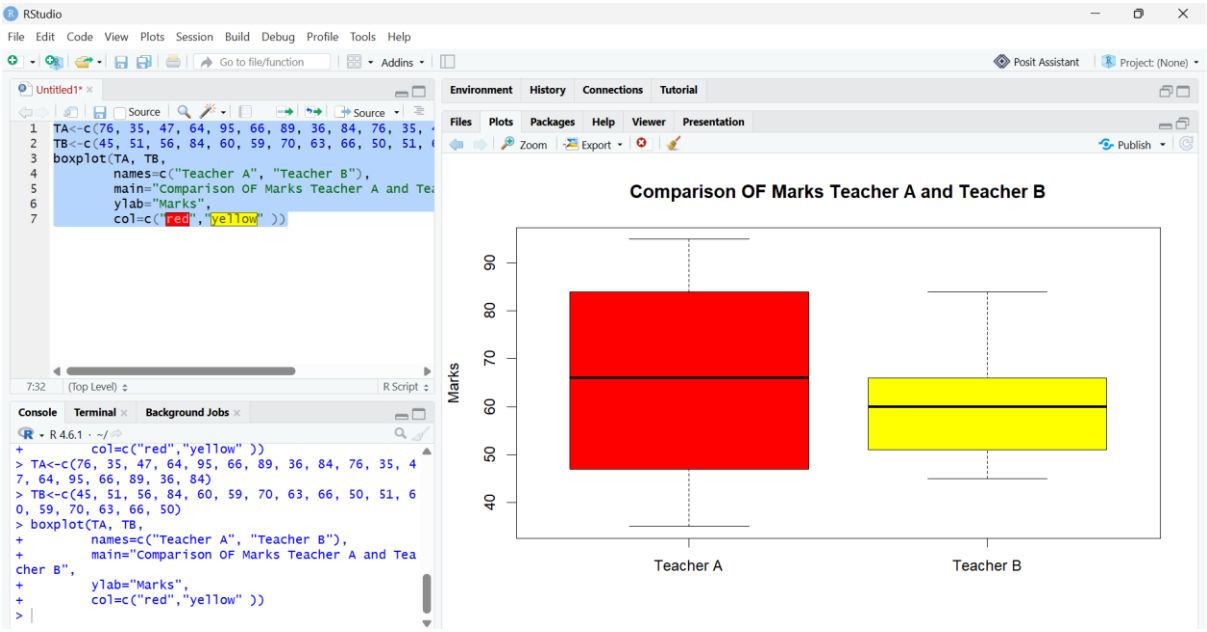
Q3.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
1 x <- c(200, 300, 400, 600, 1000)
2 min_max_norm <- (x - min(x)) / (max(x) - min(x))
3 min_max_norm
4 z_score_norm <- (x - mean(x)) / sd(x)
5 z_score_norm
Environment History Connections Tutorial
Values
min_max_norm num [1:5] 0 0.125 0.25 0.5 1
x num [1:5] 200 300 400 600 1000
z_score_norm num [1:5] -0.949 -0.632 -0.316 0.316 1.581
Console Terminal Background Jobs
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
> x <- c(200, 300, 400, 600, 1000)
> min_max_norm <- (x - min(x)) / (max(x) - min(x))
> min_max_norm
[1] 0.000 0.125 0.250 0.500 1.000
> z_score_norm <- (x - mean(x)) / sd(x)
> z_score_norm
[1] -0.9486833 -0.6324555 -0.3162278 0.3162278 1.5811388
Files Plots Packages Help Viewer Presentation
.Rhistory 2.9 KB Oct 15, 2026, 1:41 PM
Closest_Pair_of_Points_DAA_Present.pptx 50.1 KB Oct 15, 2026, 8:07 AM
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My Music
My Pictures

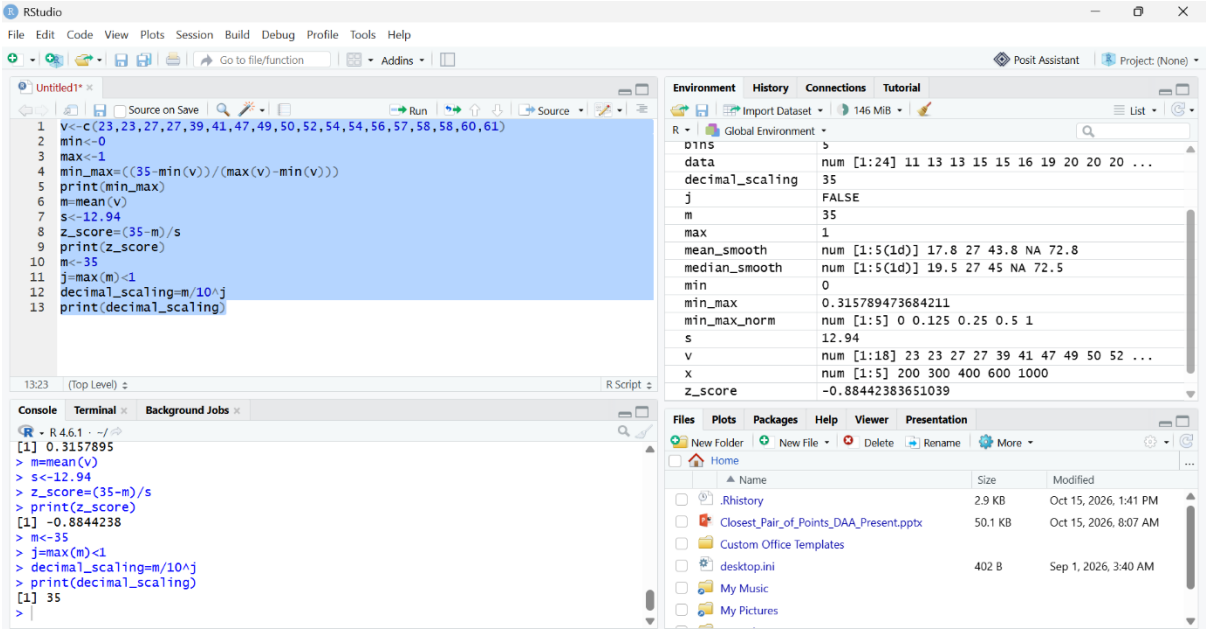
Q4.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
1 data <- c(11,13,13,15,15,16,19,20,20,20,21,21,22,23,24,30,40,45,45,45,71,72,73,7
2 bins <- 5
3 bin_indices <- cut(data, bins)
4 mean_smooth <- tapply(data, bin_indices, mean)
5 print(mean_smooth)
6 median_smooth <- tapply(data, bin_indices, median)
7 median_smooth
8 min_max_smooth <- tapply(data, bin_indices, function(x) c(min(x), max(x)))
9 print(min_max_smooth)
Environment History Connections Tutorial
Data
min_max_smooth List of 5
Values
bin_indices Factor w/ 5 levels "(10.9,23.8]",..: 1 1 1 1 1…
bins 5
data num [1:24] 11 13 13 15 15 16 19 20 20 20 ...
mean_smooth num [1:5(1d)] 17.8 27 43.8 NA 72.8
median_smooth num [1:5(1d)] 19.5 27 45 NA 72.5
min_max_norm num [1:5] 0 0.125 0.25 0.5 1
x num [1:5] 200 300 400 600 1000
z_score_norm num [1:5] -0.949 -0.632 -0.316 0.316 1.581
Console Terminal Background Jobs
$`(23.8,36.6]`
[1] 24 30
$`(36.6,49.4]`
[1] 40 45
$`(49.4,62.2]`
NULL
$`(62.2,75.1]`
[1] 71 75
Files Plots Packages Help Viewer Presentation
.Rhistory 2.9 KB Oct 15, 2026, 1:41 PM
Closest_Pair_of_Points_DAA_Present.pptx 50.1 KB Oct 15, 2026, 8:07 AM
Custom Office Templates
desktop.ini 402 B Sep 1, 2026, 3:40 AM
My Music
My Pictures

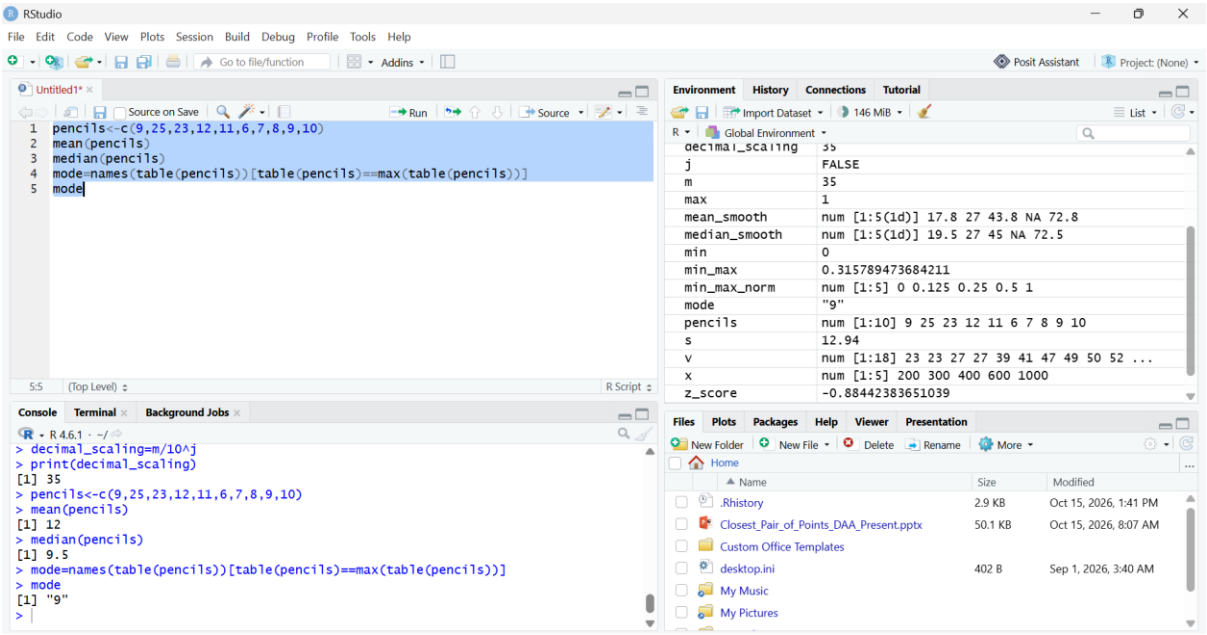
Q5.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
TA<-c(76, 35, 47, 64, 95, 66, 89, 36, 84, 76, 35,
TB<-c(45, 51, 56, 84, 60, 59, 70, 63, 66, 50, 51,
boxplot(TA, TB,
names=c("Teacher A", "Teacher B"),
main="Comparison OF Marks Teacher A and Tea
ylab="Marks",
col=c("red","yellow"))
Console Terminal Background Jobs
+ col=c("red","yellow"))
> TA<-c(76, 35, 47, 64, 95, 66, 89, 36, 84, 76, 35, 4
7, 64, 95, 66, 89, 36, 84)
> TB<-c(45, 51, 56, 84, 60, 59, 70, 63, 66, 50, 51, 6
0, 59, 70, 63, 66, 50)
> boxplot(TA, TB,
+ names=c("Teacher A", "Teacher B"),
+ main="Comparison OF Marks Teacher A and Tea
cher B",
+ ylab="Marks",
+ col=c("red","yellow"))
Comparison OF Marks Teacher A and Teacher B
Marks
40
50
60
70
80
90
Teacher A
Teacher B

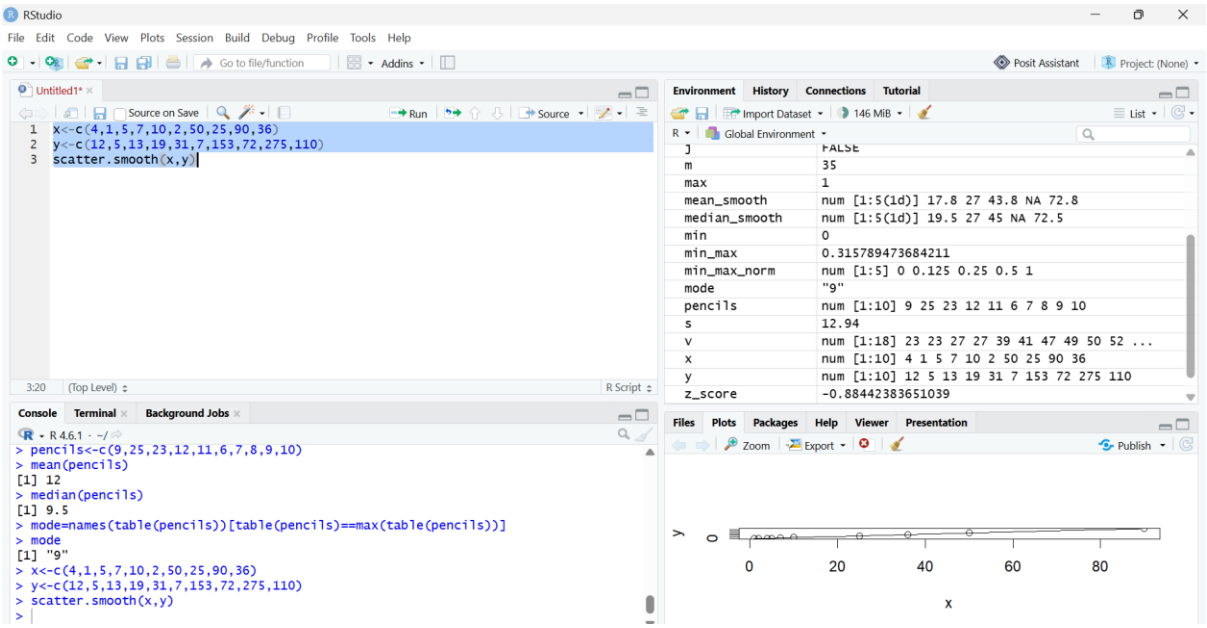
Q6.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
v<-c(23,23,27,27,39,41,47,49,50,52,54,54,56,57,58,58,60,61)
min<-0
max<-1
min_max=((35-min(v))/(max(v)-min(v)))
print(min_max)
m=mean(v)
s<-12.94
z_score=(35-m)/s
print(z_score)
m<-35
j=max(m)<1
decimal_scaling=m/10^j
print(decimal_scaling)
[1] 0.3157895
> m=mean(v)
> s<-12.94
> z_score=(35-m)/s
> print(z_score)
[1] -0.8844238
> m<-35
> j=max(m)<1
> decimal_scaling=m/10^j
> print(decimal_scaling)
[1] 35
Environment History Connections Tutorial
data num [1:24] 11 13 13 15 15 16 19 20 20 20 ...
decimal_scaling 35
j FALSE
m 35
max 1
mean_smooth num [1:5(1d)] 17.8 27 43.8 NA 72.8
median_smooth num [1:5(1d)] 19.5 27 45 NA 72.5
min 0
min_max 0.315789473684211
min_max_norm num [1:5] 0 0.125 0.25 0.5 1
s 12.94
v num [1:18] 23 23 27 27 39 41 47 49 50 52 ...
x num [1:5] 200 300 400 600 1000
z_score -0.88442383651039
Files Plots Packages Help Viewer Presentation
Home
.Rhistory 2.9 KB Oct 15, 2026, 1:41 PM
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My Music
My Pictures

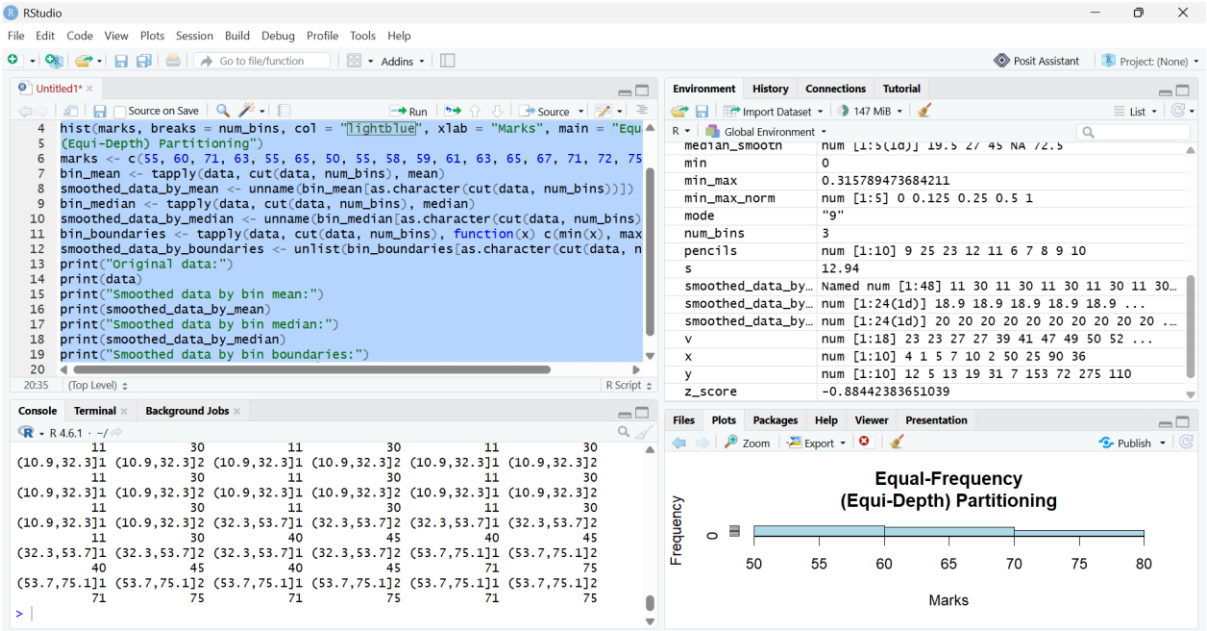
Q7.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
1 pencils<-c(9,25,23,12,11,6,7,8,9,10)
2 mean(pencils)
3 median(pencils)
4 mode=names(table(pencils))[table(pencils)==max(table(pencils))]
5 mode
Environment History Connections Tutorial
decimal_scaling 35
j FALSE
m 35
max 1
mean_smooth num [1:5(1d)] 17.8 27 43.8 NA 72.8
median_smooth num [1:5(1d)] 19.5 27 45 NA 72.5
min 0
min_max 0.315789473684211
min_max_norm num [1:5] 0 0.125 0.25 0.5 1
mode "9"
pencils num [1:10] 9 25 23 12 11 6 7 8 9 10
s 12.94
v num [1:18] 23 23 27 27 39 41 47 49 50 52 ...
x num [1:5] 200 300 400 600 1000
z_score -0.88442383651039
Console Terminal Background Jobs
> decimal_scaling=m/10^j
> print(decimal_scaling)
[1] 35
> pencils<-c(9,25,23,12,11,6,7,8,9,10)
> mean(pencils)
[1] 12
> median(pencils)
[1] 9.5
> mode=names(table(pencils))[table(pencils)==max(table(pencils))]
> mode
[1] "9"
Files Plots Packages Help Viewer Presentation
.Rhistory 2.9 KB Oct 15, 2026, 1:41 PM
Closest_Pair_of_Points_DAA_Present.pptx 50.1 KB Oct 15, 2026, 8:07 AM
Custom Office Templates
desktop.ini 402 B Sep 1, 2026, 3:40 AM
My Music
My Pictures

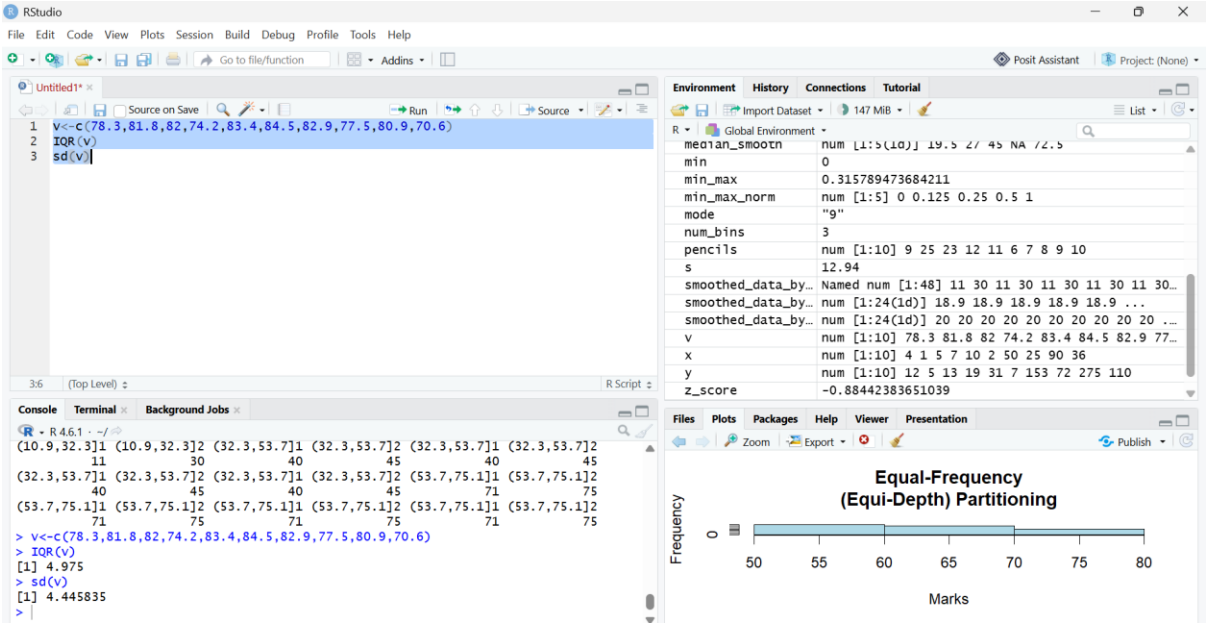
Q8.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
1 x<-c(4,1,5,7,10,2,50,25,90,36)
2 y<-c(12,5,13,19,31,7,153,72,275,110)
3 scatter.smooth(x,y)
Environment History Connections Tutorial
m 35
max 1
mean_smooth num [1:5(1d)] 17.8 27 43.8 NA 72.8
median_smooth num [1:5(1d)] 19.5 27 45 NA 72.5
min 0
min_max 0.315789473684211
min_max_norm num [1:5] 0 0.125 0.25 0.5 1
mode "9"
pencils num [1:10] 9 25 23 12 11 6 7 8 9 10
s 12.94
v num [1:18] 23 23 27 27 39 41 47 49 50 52 ...
x num [1:10] 4 1 5 7 10 2 50 25 90 36
y num [1:10] 12 5 13 19 31 7 153 72 275 110
z_score -0.88442383651039
Console Terminal Background Jobs
> pencils<-c(9,25,23,12,11,6,7,8,9,10)
> mean(pencils)
[1] 12
> median(pencils)
[1] 9.5
> mode=names(table(pencils))[table(pencils)==max(table(pencils))]
> mode
[1] "9"
> x<-c(4,1,5,7,10,2,50,25,90,36)
> y<-c(12,5,13,19,31,7,153,72,275,110)
> scatter.smooth(x,y)
Files Plots Packages Help Viewer Presentation
y
0
0 20 40 60 80
x

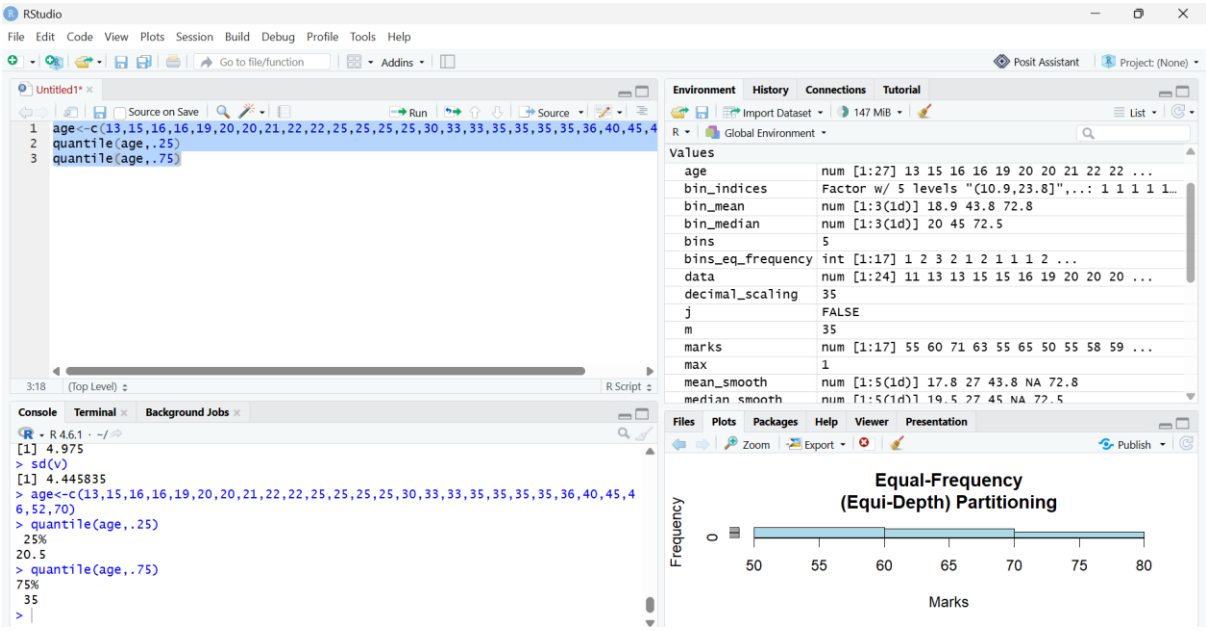
Q9.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
hist(marks, breaks = num_bins, col = "lightblue", xlab = "Marks", main = "Equ
(Equi-Depth) Partitioning")
marks <- c(55, 60, 71, 63, 55, 65, 50, 55, 58, 59, 61, 63, 65, 67, 71, 72, 75
bin_mean <- tapply(data, cut(data, num_bins), mean)
smoothed_data_by_mean <- unname(bin_mean[as.character(cut(data, num_bins))])
bin_median <- tapply(data, cut(data, num_bins), median)
smoothed_data_by_median <- unname(bin_median[as.character(cut(data, num_bins)
bin_boundaries <- tapply(data, cut(data, num_bins), function(x) c(min(x), max
smoothed_data_by_boundaries <- unlist(bin_boundaries[as.character(cut(data, n
print("Original data:")
print(data)
print("Smoothed data by bin mean:")
print(smoothed_data_by_mean)
print("Smoothed data by bin median:")
print(smoothed_data_by_median)
print("Smoothed data by bin boundaries:")
Console Terminal Background Jobs
R 4.6.1 · ~/
11 30 11 30 11 30
(10.9,32.3]1 (10.9,32.3]2 (10.9,32.3]1 (10.9,32.3]2 (10.9,32.3]1 (10.9,32.3]2
11 30 11 30 11 30
(10.9,32.3]1 (10.9,32.3]2 (10.9,32.3]1 (10.9,32.3]2 (10.9,32.3]1 (10.9,32.3]2
11 30 11 30 11 30
(10.9,32.3]1 (10.9,32.3]2 (32.3,53.7]1 (32.3,53.7]2 (32.3,53.7]1 (32.3,53.7]2
11 30 40 45 40 45
(32.3,53.7]1 (32.3,53.7]2 (32.3,53.7]1 (32.3,53.7]2 (53.7,75.1]1 (53.7,75.1]2
40 45 40 45 71 75
(53.7,75.1]1 (53.7,75.1]2 (53.7,75.1]1 (53.7,75.1]2 (53.7,75.1]1 (53.7,75.1]2
71 75 71 75 71 75
Environment History Connections Tutorial
Import Dataset 147 MiB
Global Environment
median_smooth num [1:5(1d)] 19.5 27 45 NA 72.5
min 0
min_max 0.315789473684211
min_max_norm num [1:5] 0 0.125 0.25 0.5 1
mode "9"
num_bins 3
pencils num [1:10] 9 25 23 12 11 6 7 8 9 10
s 12.94
smoothed_data_by... Named num [1:48] 11 30 11 30 11 30 11 30 11 30...
smoothed_data_by... num [1:24(1d)] 18.9 18.9 18.9 18.9 18.9 ...
smoothed_data_by... num [1:24(1d)] 20 20 20 20 20 20 20 20 20 20 ...
v num [1:18] 23 23 27 27 39 41 47 49 50 52 ...
x num [1:10] 4 1 5 7 10 2 50 25 90 36
y num [1:10] 12 5 13 19 31 7 153 72 275 110
z_score -0.88442383651039
Files Plots Packages Help Viewer Presentation
Zoom Export Publish
Equal-Frequency
(Equi-Depth) Partitioning
Frequency
0
50 55 60 65 70 75 80
Marks

Q10.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
v<-c(78.3,81.8,82,74.2,83.4,84.5,82.9,77.5,80.9,70.6)
IQR(v)
sd(v)
Console Terminal Background Jobs
R 4.6.1 · ~/
(10.9,32.3]1 (10.9,32.3]2 (32.3,53.7]1 (32.3,53.7]2 (32.3,53.7]1 (32.3,53.7]2
11 30 40 45 40 45
(32.3,53.7]1 (32.3,53.7]2 (32.3,53.7]1 (32.3,53.7]2 (53.7,75.1]1 (53.7,75.1]2
40 45 40 45 71 75
(53.7,75.1]1 (53.7,75.1]2 (53.7,75.1]1 (53.7,75.1]2 (53.7,75.1]1 (53.7,75.1]2
71 75 71 75 71 75
> v<-c(78.3,81.8,82,74.2,83.4,84.5,82.9,77.5,80.9,70.6)
> IQR(v)
[1] 4.975
> sd(v)
[1] 4.445835
Environment History Connections Tutorial
Import Dataset 147 MiB
Global Environment
median_smooth num [1:5(1d)] 19.5 27 45 NA 72.5
min 0
min_max 0.315789473684211
min_max_norm num [1:5] 0 0.125 0.25 0.5 1
mode "9"
num_bins 3
pencils num [1:10] 9 25 23 12 11 6 7 8 9 10
s 12.94
smoothed_data_by... Named num [1:48] 11 30 11 30 11 30 11 30 11 30...
smoothed_data_by... num [1:24(1d)] 18.9 18.9 18.9 18.9 18.9 ...
smoothed_data_by... num [1:24(1d)] 20 20 20 20 20 20 20 20 20 20 ...
v num [1:10] 78.3 81.8 82 74.2 83.4 84.5 82.9 77...
x num [1:10] 4 1 5 7 10 2 50 25 90 36
y num [1:10] 12 5 13 19 31 7 153 72 275 110
z_score -0.88442383651039
Files Plots Packages Help Viewer Presentation
Zoom Export Publish
Equal-Frequency
(Equi-Depth) Partitioning
Frequency
0
50 55 60 65 70 75 80
Marks

Q11.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled1*
1 age<-c(13,15,16,16,19,20,20,21,22,22,25,25,25,25,30,33,33,35,35,35,35,36,40,45,4
2 quantile(age,.25)
3 quantile(age,.75)
Console Terminal Background Jobs
[1] 4.975
> sd(v)
[1] 4.445835
> age<-c(13,15,16,16,19,20,20,21,22,22,25,25,25,25,30,33,33,35,35,35,35,36,40,45,4
6,52,70)
> quantile(age,.25)
25%
20.5
> quantile(age,.75)
75%
35
Environment History Connections Tutorial
Values
age num [1:27] 13 15 16 16 19 20 20 21 22 22 ...
bin_indices Factor w/ 5 levels "(10.9,23.8]",...: 1 1 1 1 1...
bin_mean num [1:3(1d)] 18.9 43.8 72.8
bin_median num [1:3(1d)] 20 45 72.5
bins 5
bins_eq_frequency int [1:17] 1 2 3 2 1 2 1 1 1 2 ...
data num [1:24] 11 13 13 15 15 16 19 20 20 20 ...
decimal_scaling 35
j FALSE
m 35
marks num [1:17] 55 60 71 63 55 65 50 55 58 59 ...
max 1
mean_smooth num [1:5(1d)] 17.8 27 43.8 NA 72.8
Files Plots Packages Help Viewer Presentation
Equal-Frequency
(Equi-Depth) Partitioning
Frequency
0
50 55 60 65 70 75 80
Marks